Set 1: Scientific notation and unit conversions

Exponential notation

This notation, also known as scientific notation, is used for convenience when writing very large or very small numbers.

To express a number using experimental notation, it is written as a number between 1 and 10 multiplied by the appropriate power of 10.

For examples 0.056 expressed in exponential notation is written as 5.6×10^{-2} while 167,000 would be written as 1.67×10^5.

Arithmetic using exponential notation is governed by the following rules:

Addition and subtraction Before adding or subtracting, numbers must be expressed to the same powers of 10.	**Multiplication** When multiplying powers of 10 add their indices algebraically.	**Division** When dividing powers of 10 by each other, subtract the index of the denominator from that of the numerator.
Example: $(2.04 \times 10^5) + (4.7 \times 10^4)$ $= (2.04 \times 10^5) + (0.47 \times 10^5)$ $= (2.04 + 0.47) \times 10^5$ $= 2.51 \times 10^5$	Example: $(5 \times 10^5) \times (4 \times 10^2)$ $= (5 \times 4) \times 10^{5+2}$ $= 20 \times 10^7$ $= 2 \times 10^8$	Examples: $\frac{1.8 \times 10^8}{6 \times 10^5} = \frac{1.8 \times 10^{8-5}}{6}$ $= 0.3 \times 10^3$ $= 3 \times 10^2$

Notes

Set 1: Exercises

1. Express the following numbers using exponential notation:

(a) 329
(b) 1006
(c) 0.5731
(d) 67 240 000 000
(e) 0.04
(f) 0.000 000 078

2. Convert the following measurements into the required units:

(a) 2.643 kg to g
(b) 0.012 kg to g
(c) 2.50 tonne to g
(d) 439 mg to g
(e) 0.2846 g to mg
(f) 6.72 × 108 mg to kg

3. Complete the following:

(a) 10.2 m = ? cm
(b) 1.26 cm = ? m
(c) 1.46 m = ? mm
(d) 143 267 mm = ? m
(e) 109 nm = ? m
(f) 141 mm^2 = ? m^2
(h) 8.3 mm^3 = ? m^3
(i) 1220 cm = ? m
(j) 15 mm = ? cm
(k) 10 dm = ? m
(1) 1.9 µm = ? cm
(m) 15 cm^2 = ? mm^2
(o) 4.9 cm^3 = ? m^3
(p) 1.67 mL = ? L